

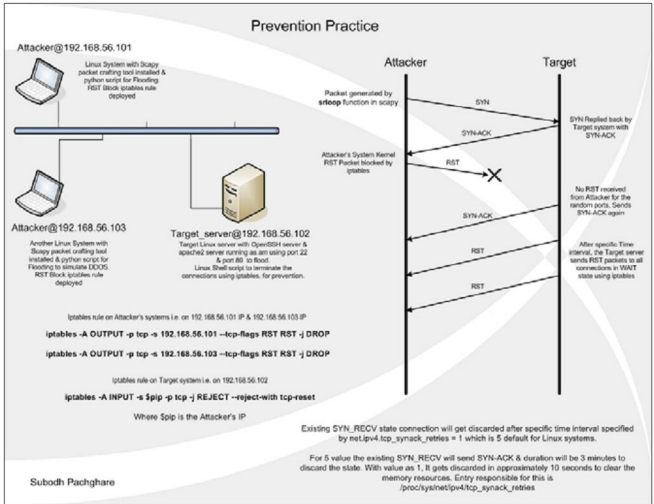


Figure 3: Prevention practice

The EXIT signal (^C) will cause the script to flush all *iptables* chains, and clear all the rules it added. The topology diagram is in Figure 3. Figure 2 shows the Scapy SYN packet output at the shell prompt.

Kernel configuration

Beyond this shell script, the number of SYN_ACK retries can also be lowered, so that the kernel closes SYN_RECV state connections earlier. The parameter *net.ipv4.tcp_synack_retries* defaults to 5 SYN_ACK retries, which leaves SYN_RECV-state connections open for 3 minutes. You can reduce this so that these hanging connections will close sooner. I have used a value of 1, just for demonstration. This allows a time interval of 10 seconds before the connection is closed. Set the parameter using the following method. Add or edit the following line in */etc/sysctl.conf*:

```
net.ipv4.tcp_synack_retries = 1
```

Commit the changes made using *sysctl*:

```
root@ubiserv:~# sysctl -p /etc/sysctl.conf
```

Verify it using the following code:

```
root@ubiserv:~# cat /proc/sys/net/ipv4/tcp_synack_retries
1
```

So, this is what you need to know to prevent a SYN queue flood. Please drop me a mail if you are thinking about using this, and of course, for any suggestions or problems. **END** 

By: Subodh V Pachghare

The author works with India's largest commercial automotive organisation, in the IT division. He is interested in the networking, information security, back-up and virtualisation domains. He is a FOSS follower, and spends most of his time learning the internals of Linux. You can reach him at www.subodh.com or email him at subodhpachghare@gmail.com